

# Task 3 | Boundary-aware audit of a retraction-associated FASTQ record

A

## Boundary-stress question

Can a high-risk literature record be converted into a bounded source-data audit?



### Public accession

SRP508771 / PRJNA1111407 / SRR29088443

### Predefined boundary

verify files and QC outputs;  
do not rescue the biological claim

B

## Published record vs agent output

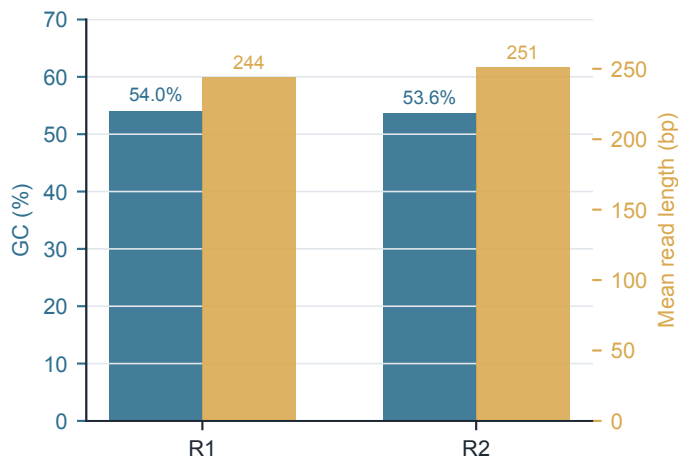
Direct source-data comparison shows exact file-level reproducibility.

| Metric       | Public     | Agent      | Difference |
|--------------|------------|------------|------------|
| R1 reads     | 53,571     | 53,571     | 0          |
| R2 reads     | 53,571     | 53,571     | 0          |
| Total bases  | 26,517,645 | 26,517,645 | 0          |
| R1 file size | 1,240,071  | 1,240,071  | 0          |
| R2 file size | 1,428,410  | 1,428,410  | 0          |

All five source-level checks matched exactly

C

## QC outputs generated from the same files



Malformed records: 0; max N fraction: 4e-06

D

## What the agent demonstrates

The output is useful because it is bounded, not because it overclaims.

### CAN verify

- file identity
- paired-end structure
- read/base/file metadata
- QC report provenance

### CANNOT claim

- retracted conclusion
- ancient authenticity
- contamination source
- biological mechanism

Result: source-data reproducibility = PASS;  
claim-level interpretation = expert-gated